

AKCMamba-YOLO: Selective State Space Models For Real-Time Object Detection

Long Chen¹, Hui Wang¹, Man Xu², Zexuan Li², Zizhu Fan^{2*}

¹East China Jiaotong University, Nanchang, China

xlllchen@ecjtu.edu.cn, huiwangens@163.com

²Shanghai University of Electric Power, Shanghai, China

xumanly@shiep.edu.cn, lizexuan@shiep.edu.cn, zzfan3@shiep.edu.cn

Abstract

The You Only Look Once (YOLO) series has been a cornerstone in real-time object detection, renowned for its efficient convolutional design and rapid inference. However, its reliance on convolutional operations inherently limits its ability to capture long-range dependencies and rich contextual information, leading to suboptimal performance in complex scenes. Recently, State Space Models (SSM) have emerged as an efficient alternative to attention mechanisms, offering global representation with linear time complexity. In this paper, we propose AKCMamba-YOLO, a novel object detector that incorporates SSM into the YOLO architecture. We introduce 3CAKCMamba and 4CAKCMamba modules to a novel object detection framework, enabling enhanced channel interaction and cross-layer semantic fusion. This design improves multi-scale feature modeling while maintaining computational efficiency. To support safety-critical applications, we provide railway pedestrian detection dataset with 2,975 annotated images under complex scenarios. Experiments on COCO2017, Power Tower Foreign Object Detection Datasets, and our custom dataset show that AKCMamba-YOLO achieves superior accuracy and speed compared to state-of-the-art baselines, making it well-suited for real-time and resource-constrained environments. Code is available at https://github.com/xlllchen/AKCMamba_YOLO

1. Introduction

Object detection, a cornerstone of computer vision, has been profoundly advanced by deep learning architectures. Among these, the YOLO series has established the paradigm for real-time detection, skillfully balancing accuracy and efficiency through successive innovations from

YOLOv4 [26] to YOLOv8 [14] and YOLOv11 [15]. Despite their prowess, these models predominantly rely on convolutional operations, which inherently possess a local receptive field [34]. This fundamental characteristic limits their ability to model long-range dependencies and rich contextual information, leading to suboptimal performance in complex scenarios involving multi-scale objects, significant occlusion, or scenes requiring global reasoning.

The Transformer architecture [28], with its powerful self-attention mechanism, emerged as a compelling solution for capturing global context [4]. However, its adoption in real-time detection systems is hampered by the quadratic computational complexity of self-attention with respect to input size, resulting in substantial computational overhead and latency [23]. This computational inefficiency underscores a critical challenge: the need for an architectural backbone that retains the high-speed, low-complexity profile of convolutional networks while possessing the global representational capacity of Transformers.

Recently, SSM like Mamba [7, 36] have surfaced as transformative alternative. By leveraging an input-dependent selective mechanism and a linear-time recurrent formulation [2], SSM achieve efficient long-sequence modeling [42], demonstrating remarkable success in language and image classification tasks [21]. This breakthrough prompts a pivotal research question: Can the selective state space models be effectively integrated into the YOLO framework to bridge the capability gap, endowing it with global contextual awareness without compromising its real-time performance?

In this paper, we present AKCMamba-YOLO, a novel object detection framework, deeply integrates selective state space modeling with the YOLO architecture. Our core innovation lies in the design of two content-aware modules: the 3CAKCMamba module for the backbone and the 4CAKCMamba module for the neck. These modules replace the traditional C2f blocks, leveraging a founda-

*Corresponding author.

tional adaptive kernel convolution block (AKCBlock) that incorporates dynamic kernel convolution [39] to enable a paradigm shift from static, local feature extraction to dynamic, sequential modeling. This design facilitates enhanced channel interaction, deep cross-layer semantic fusion, and adaptive multi-scale feature integration.

To support research and deployment in safety-critical domains, we further contribute a dedicated Railway Pedestrian Detection Dataset, comprising 2,975 meticulously annotated images under diverse and challenging scenarios. Extensive experiments conducted on COCO2017 [20], Power Tower Foreign Object Dataset [40], and our Railway Pedestrian Dataset demonstrate the superior performance of AKCMamba-YOLO over state-of-the-art baselines.

The main contributions of this paper are summarized as follows:

- Introduces a object detection framework that deeply integrates selective state space models with the YOLO architecture, resolving the fundamental conflict between long-range dependency modeling and computational efficiency in traditional methods.
- Designed content-aware 3CAKCMamba and 4CAKCMamba modules, achieving a paradigm shift in feature extraction from static convolution to dynamic sequence modeling.
- Established a railway pedestrian detection benchmark for safety-critical applications, providing the research community with an important evaluation resource.

2. Related Work

2.1. Evolution of YOLO Series

The YOLO series has consistently advanced real-time object detection through architectural innovations. YOLOv4 introduced CSPDarknet53 with cross stage partial connections to enhance gradient flow [3]. Subsequent versions refined this foundation: YOLOv7 employed E-ELAN for better feature reuse [30], while YOLOv8 streamlined design with C2f modules and anchor-free heads. Recent developments include YOLOv9’s GELAN structure [31], YOLOv10’s NMS-free strategy [28], and YOLOv11’s edge optimization [15].

Despite these advances, the series remains limited by convolutional operations’ constrained receptive fields and weak contextual modeling, especially for small or occluded objects. While attention-enhanced variants like Gold-YOLO [43] address some limitations, they incur computational overhead from dense attention mechanisms. Our approach integrates state space modeling to enhance global context capture while maintaining real-time efficiency for small or occluded objects.

2.2. End-to-End Object Detectors

End-to-end detectors have revolutionized object detection by eliminating handcrafted components like anchor generation and NMS. DETR [4] pioneered this approach using Transformer architectures for direct set prediction, though it suffered from slow convergence and poor small-object performance. Subsequent improvements include deformable DETR’s sparse sampling [44], DINO’s denoising training strategy [37], and RT-DETR’s hybrid encoder design for real-time performance [17]. Despite these advances, Transformer-based detectors face challenges with computational overhead and complex training dynamics, especially in lightweight configurations. In contrast, YOLO-based detectors maintain advantages in practical deployment due to their lightweight design and efficient multi-scale feature aggregation, making them ideal foundations for integrating emerging paradigms like state space models in efficiency-critical applications.

2.3. State Space Models in Visual Recognition

SSM have emerged as competitive alternatives to Transformers, offering linear complexity and strong sequence modeling capabilities [9]. While S4 [8] and DSS [13] established structured state spaces, Mamba [7] introduced selective state updates for adaptive context integration. This progress has spurred SSM adoption in vision tasks: Vision Mamba [36] and VMamba [22] developed pure SSM backbones and 2D spatial modeling, while LocalMamba [12] balanced global and local context through windowed scanning. Although studies like MambaOut question SSM’s value in static classification, their advantages in memory efficiency, dynamic receptive fields, and continuity modeling remain promising for dense prediction tasks, as demonstrated in medical imaging [16] and remote sensing [1].

Most related to our work is Mamba YOLO [32], which integrates SSM into YOLO’s backbone. AKCMamba-YOLO advances this by deeper integration: we replace both backbone and neck with specialized 3CAKCMamba and 4CAKCMamba blocks, incorporate adaptive kernel convolution (AKConv) [39] for adaptive feature extraction, and achieve more comprehensive cross-scale fusion. This holistic approach enables superior multi-scale modeling while maintaining the efficiency required for real-time applications.

3. Method

3.1. Preliminaries

3.1.1 AKConv Module

To overcome the limitations of standard convolution in handling variably shaped targets, we adopt AKConv, which introduces kernels with learnable sampling shapes and lin-

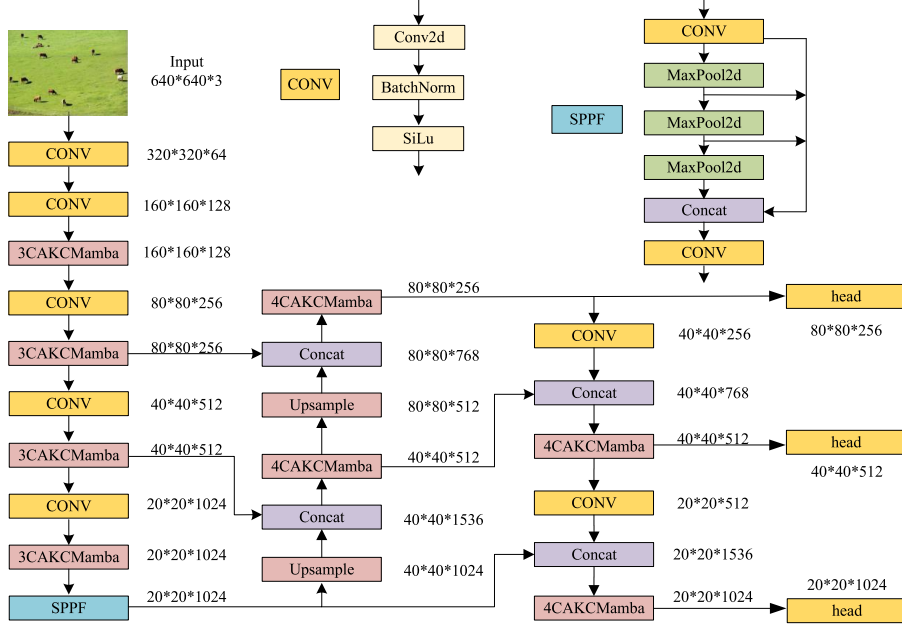


Figure 1. Overview of AKCMamba-YOLO. 3CAKCMamba enhances channel interaction and cross-layer semantic fusion; 4CAKCMamba aggregates multi-scale features for robust spatial-channel representation.

ear parameter growth. Unlike traditional convolution—limited by fixed sampling patterns and quadratic parameter increase—AKConv adaptively adjusts sampling locations via a set of learnable offsets:

$$\hat{P}_n = P_n + \Delta P_n, \quad (1)$$

where P_n denotes initial coordinates and ΔP_n the learned offsets. The convolution at position p_0 is computed as:

$$\text{AKConv}(p_0) = \sum_{n=1}^N w_n \cdot X(p_0 + \hat{P}_n). \quad (2)$$

This adaptive sampling mechanism captures irregular structures more effectively while maintaining linear parameter growth. By integrating AKConv in place of standard convolution, we achieve a favorable balance between representation capacity and computational cost, making it suitable for tasks involving diverse geometric forms.

3.1.2 State Space Models

The Mamba model [32] employs a simplified block architecture that integrates features from H3 and GatedMLP blocks, enhancing both computational efficiency and expressive power through SSM and advanced activation functions.

At the core of Mamba lies the SSM, which maps a 1D sequence $x(t) \in \mathbb{R}$ to an output $y(t) \in \mathbb{R}$ via a latent state

$h(t) \in \mathbb{R}^N$. The continuous-time SSM is defined as:

$$h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t), \quad y(t) = \mathbf{C}h(t), \quad (3)$$

where $\mathbf{A} \in \mathbb{R}^{N \times N}$ is the state matrix, and $\mathbf{B} \in \mathbb{R}^{N \times 1}$, $\mathbf{C} \in \mathbb{R}^{1 \times N}$ are projection parameters.

For discrete-time inputs, the system is transformed using a timescale parameter Δ :

$$h_k = \bar{\mathbf{A}}h_{k-1} + \bar{\mathbf{B}}x_k, \quad y_k = \mathbf{C}h_k, \quad (4)$$

where $\bar{\mathbf{A}} = \exp(\Delta\mathbf{A})$ and $\bar{\mathbf{B}} = (\Delta\mathbf{A})^{-1}(\exp(\Delta\mathbf{A}) - \mathbf{I}) \cdot \Delta\mathbf{B}$ under a zero-order hold (ZOH) discretization.

A key innovation in Mamba is the selective mechanism, where $\mathbf{B}$, $\mathbf{C}$, and Δ become input-dependent, allowing context-aware filtering. The output can also be computed via global convolution:

$$y = x * \bar{\mathbf{K}}, \quad \bar{\mathbf{K}} = (\mathbf{C}\bar{\mathbf{B}}, \mathbf{C}\bar{\mathbf{A}}\bar{\mathbf{B}}, \dots, \mathbf{C}\bar{\mathbf{A}}^{N-1}\bar{\mathbf{B}}), \quad (5)$$

where $\bar{\mathbf{K}} \in \mathbb{R}^N$ forms a structured convolutional kernel.

3.2. Overall Architecture

The overall architecture of AKCMamba-YOLO is depicted in Figure 1. Built upon the YOLOv8 framework, our approach introduces significant enhancements through the strategic integration of state space models and adaptive convolution mechanisms. The key innovation lies in the systematic replacement of traditional components with specialized modules designed to address the limitations of convolutional operations in modeling long-range dependencies.

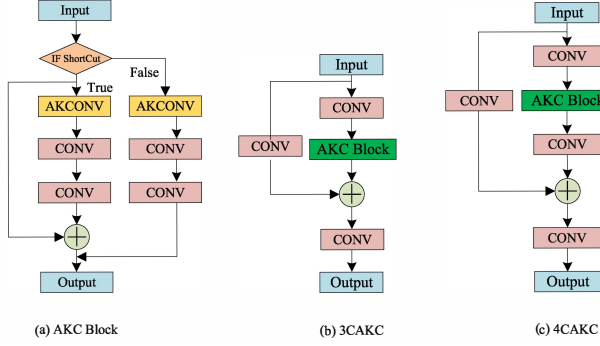


Figure 2. (a) Illustration of AKCBlock architecture; (b) Illustration of 3CAKC architecture; (c) Illustration of 4CAKC architecture.

For the backbone network, we substitute the conventional C2f modules with our proposed 3CAKCMamba modules. This replacement enables more effective feature extraction through a three-layer structure that combines adaptive kernel convolution with state space modeling. The 3CAKCMamba module facilitates deep feature mining and cross-layer semantic fusion while maintaining computational efficiency.

For the neck network, we introduce 4CAKCMamba modules to replace the original C2f modules. These four-layer structures enhance feature map channels and introduce advanced interaction mechanisms to better capture multi-scale and contextual information. The hierarchical design supports efficient feature integration and reorganization, significantly improving target localization and classification performance in the detection head.

Beyond these core replacements, AKCMamba-YOLO incorporates several foundational components: the AKCBlock as the basic building unit, AKSS2D for spatial-state space modeling, and AKCAttention for adaptive feature recalibration. Together, these innovations form a cohesive framework that balances detection accuracy with computational efficiency.

3.3. Core Module

3.3.1 AKCBlock: Adaptive Kernel Convolution Block

The AKCBlock serves as the fundamental unit in our architecture, incorporating a dynamic shortcut mechanism that adaptively switches between residual connections and direct outputs. As illustrated in Figure 2(a), this design balances network flexibility and training stability [6]. Formally, given input feature maps z^{l-1} , the AKCBlock operation is defined as:

$$z^l = \begin{cases} \omega(\omega(AKCONV(z^{l-1}))) \oplus z^{l-1} & \text{if true} \\ \omega(\omega(AKCONV(z^{l-1}))) & \text{if false} \end{cases} \quad (6)$$

where ω denotes 1×1 convolution for channel alignment, and $\oplus$ represents element-wise addition. The AKCONV operation [39] enables dynamic kernel adaptation, allowing the model to adjust receptive fields based on input characteristics.

3.3.2 3CAKC and 4CAKC: Multi-Scale Feature Extraction

Building upon the AKCBlock, we design hierarchical feature extraction modules that efficiently capture multi-scale representations. The 3CAKC module, shown in Figure 2(b), follows a streamlined processing pipeline:

$$z^l = \omega(AKC(\omega(z^{l-1}))) \oplus \omega(z^{l-1}). \quad (7)$$

The 4CAKC module, depicted in Figure 2(c), extends this design with an additional convolutional layer for enhanced feature transformation:

$$z^l = \omega(\omega(AKC(\omega(z^{l-1})))) \oplus \omega(z^{l-1}). \quad (8)$$

These modules effectively address the feature hierarchy in deep networks while mitigating gradient vanishing issues through residual connections.

3.3.3 AKSS2D: Long-range dependency modeling

The AKSS2D module integrates adaptive convolution with state space modeling to capture long-range spatial dependencies [9]. As shown in Figure 3(a), the module operates through three key operations:

Scanning Expansion: The input feature map is unfolded into sequences along four directional scans (top-left to bottom-right, bottom-left to top-right, bottom-right to top-left, and top-right to bottom-left), ensuring comprehensive spatial coverage.

S6 Block Processing: Each sequence undergoes processing through the S6 block [7], which implements selective state space modeling with input-dependent parameter adaptation.

Scanning Merging: The processed sequences from all directions are summed and merged, then reshaped to restore the original spatial dimensions.

The AKSS2D formulation incorporates AKCONV for initial feature adaptation, as shown in Figure 3(a):

$$z^l = \text{LN}(\text{AKCONV}(z^{l-1})), \quad (9)$$

where LN denotes linear transformation for feature projection.

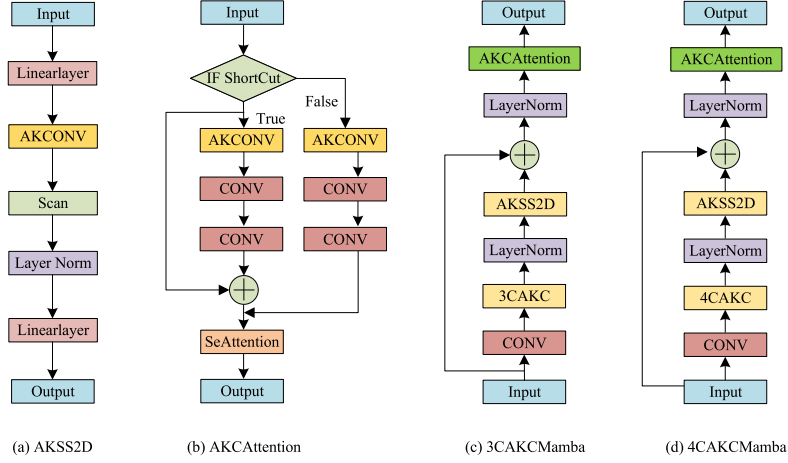


Figure 3. Overview of the proposed modules. (a) AKSS2D for long-range spatial modeling; (b) AKCAAttention for adaptive feature recalibration; (c) 3CAKCMamba, and (d) 4CAKCMamba.

3.3.4 AKCAAttention: Adaptive Feature Recalibration

The AKCAAttention module, illustrated in Figure 3(b), enhances critical features while suppressing redundant information through spatial-channel attention mechanisms [10, 19]. The module operates as:

$$z^l = \begin{cases} SeA(\omega(\omega(AKCONV(z^{l-1}))) \oplus z^{l-1}) & \text{if true} \\ SeA(\omega(\omega(AKCONV(z^{l-1})))) & \text{if false,} \end{cases} \quad (10)$$

where SeA represents the squeeze-and-excitation attention operation that recalibrates feature importance based on inter-channel dependencies.

3.3.5 3CAKCMamba and 4CAKCMamba: Integrated Detection Modules

The 3CAKCMamba and 4CAKCMamba modules represent the complete integration of our proposed components into cohesive detection units. As shown in Figure 3, the 3CAKCMamba processing flow is formalized as:

$$z^l = \psi(\text{LN}(\phi(\text{LN}(3\text{CAKC}(\omega(z^{l-1})))) \oplus \omega(z^{l-1}))), \quad (11)$$

where ψ denotes AKCAAttention and ϕ represents AKSS2D. Similarly, the 4CAKCMamba module follows:

$$z^l = \psi(\text{LN}(\phi(\text{LN}(4\text{CAKC}(\omega(z^{l-1})))) \oplus \omega(z^{l-1}))). \quad (12)$$

These modules systematically combine local feature extraction (3CAKC/4CAKC), long-range dependency modeling (AKSS2D), and adaptive feature selection (AKCAAttention) to form comprehensive detection blocks.

4. Experiments

4.1. Experimental Setup

4.1.1 Datasets and Evaluation Metrics

We conduct comprehensive evaluations on three challenging datasets to validate the effectiveness and generalizability of AKCMamba-YOLO.

COCO2017 Dataset [20]. We use the standard train2017 split (118K images) for training and val2017 split (5K images) for validation, following established evaluation protocols for general object detection.

Power Tower Foreign Object Detection Dataset [40]. A specialized industrial dataset containing 2,425 training and 500 validation images for detecting safety-critical anomalies around power infrastructure, including birds, drones, kites, and other foreign objects.

Railway Pedestrian Detection Dataset. Our newly introduced safety-critical benchmark comprising 2,975 training and 600 validation images captured under diverse railway scenarios, addressing the challenging task of pedestrian detection in complex railway environments.

All models are evaluated using standard object detection metrics: mean Average Precision (mAP), AP50, AP75, and AP50:95. For efficiency analysis, we report parameters, FLOPs, and FPS on an NVIDIA V100 GPU.

4.1.2 Implementation Details

We implement AKCMamba-YOLO based on YOLOv8, training all models for 500 epochs with batch size 32 using SGD optimizer. Data augmentation includes Mosaic (p=1.0) and HSV transformations (hue: 0.015, saturation: 0.7, value: 0.4). Input size is fixed at 640×640 for both training and inference. The loss function uses weights of

7.5 for box loss, 0.5 for cls loss, and 1.5 for DFL loss. We employ a constant learning rate of 0.01 after a 3-epoch warm-up (bias lr: 0.1, momentum: 0.8), with weight decay of 0.0005.

4.2. Main Results

4.2.1 Comparison on COCO2017 Dataset

Table 1 presents comprehensive comparisons with state-of-the-art detectors on the COCO2017 val set. AKCMamba-YOLO demonstrates superior performance across multiple dimensions.

Table 1. The comparison of AKCMamba-YOLO with other detectors from the YOLO series on the COCO2017 val.

Model	Accuracy (%)			Params	FLOPs
	mAP	AP50	AP75		
YOLOv5-N [25]	28.0	45.7	-	1.9M	4.5G
YOLOv6-N [18]	37.0	52.7	-	4.7M	4.7G
YOLOv7 [30]	37.4	55.2	37.3	6.2M	13.7G
YOLOv8-N [14]	37.3	52.6	40.6	3.2M	8.7G
YOLOv8-S [14]	44.9	61.8	48.6	11.2M	28.6G
DAMO YOLO-S [35]	46.0	61.9	49.5	12.3M	37.8G
Gold-YOLO-N [29]	39.6	55.7	-	5.6M	12.1G
YOLO MS-X [27]	43.4	60.4	47.6	4.5M	17.4G
Mamba YOLO-T [32]	45.4	62.3	49.1	6.1M	14.3G
OURS	46.3	63.1	51.4	9.1M	14.9G

Our method achieves state-of-the-art performance with 46.3% mAP, outperforming YOLOv8-S by 1.4% absolute improvement while reducing FLOPs by 47.9%. Compared to the recent Mamba YOLO-T, we achieve improvements of +0.9% mAP, +0.8% AP50, and +2.3% AP75, demonstrating the effectiveness of our comprehensive architectural integration.

4.2.2 Power Tower Foreign Object Detection Dataset

For industrial safety applications, we evaluate our method on the challenging Power Tower Foreign Object Detection Dataset. As shown in Table 2, AKCMamba-YOLO achieves remarkable performance gains in this specialized domain.

Our method achieves 92.8% precision with 86.9% AP50 and 72.5% AP50:95, representing significant improvements over YOLOv8-S (+2.5% precision, +3.0% AP50, +2.4% AP50:95). The robust performance in detecting small foreign objects like bird nests and kite strings demonstrates the practical value of our approach for industrial safety monitoring.

4.2.3 Railway Pedestrian Detection Dataset

Table 3 presents results on our Railway Pedestrian Dataset, highlighting AKCMamba-YOLO’s effectiveness in safety-critical scenarios.

Table 2. Performance obtained by the competing methods on the Power Tower Foreign Object Detection Dataset.

Model	Metrics (%)			FLOPs
	Precision	AP50	AP50:95	
YOLOv5-N [25]	83.4	83.2	68.4	4.5G
YOLOv6-N [18]	85.3	84.5	69.5	4.7G
YOLOv7-Tiny [30]	89.4	85.6	70.3	13.7G
YOLOv8-N [14]	89.7	83.3	67.4	8.7G
YOLOv8-S [14]	90.3	83.9	70.1	28.6G
YOLOv10 [28]	91.5	85.4	71.3	25G
YOLOv11 [15]	92.3	86.1	71.8	40G
DAMO YOLO-S [35]	91.4	85.8	71.5	37.8G
Gold-YOLO-N [29]	90.3	82.9	68.1	12.1G
YOLO MS-X [27]	92.0	86.1	71.2	17.4G
Mamba YOLO-T [32]	92.1	86.3	71.3	14.3G
OURS	92.8	86.9	72.5	14.9G

Table 3. Performance obtained by the competing methods on the Railway Pedestrian Detection Dataset.

Model	Metrics (%)			FLOPs
	Precision	AP50	AP50:95	
YOLOv5-N [25]	83.4	85.2	66.5	4.5G
YOLOv6-N [18]	89.1	90.5	69.4	4.7G
YOLOv7-Tiny [30]	93.5	94.2	73.1	13.7G
YOLOv8-N [14]	93.2	95.1	73.7	8.7G
YOLOv8-S [14]	94.6	97.2	74.2	28.6G
YOLOv10 [28]	94.3	96.8	74.7	25G
YOLOv11 [15]	94.8	97.1	75.1	40G
DAMO YOLO-S [35]	94.1	96.3	73.8	37.8G
Gold-YOLO-N [29]	94.3	96.6	74.3	12.1G
YOLO MS-X [27]	94.8	96.9	74.6	17.4G
Mamba YOLO-T [32]	94.8	97.1	75.1	14.3G
OURS	95.1	97.4	75.5	14.9G

Our method achieves 95.1% precision with 97.4% AP50 and 75.5% AP50:95, showing consistent improvements over baseline methods. While the absolute gains may appear modest, in safety-critical railway applications where every detection matters, these improvements translate to enhanced operational safety and reliability.

4.3. Ablation Studies

To systematically evaluate the contribution of each proposed component in AKCMamba-YOLO, we conduct comprehensive ablation studies on the Power Tower Foreign Object Dataset. All experiments maintain identical training/validation splits, training epochs, and post-processing procedures to ensure fair comparisons.

4.3.1 Backbone Component Analysis

We first investigate the impact of individual components in the backbone network by progressively integrating our proposed modules. As shown in Table 4, each component contributes significantly to the overall performance improvement.

The baseline model using standard C2f modules achieves

Table 4. Ablation study on backbone components using Power Tower Foreign Object Dataset.

3CAKC	AKSS2D	AKCAttention	Precision(%)	AP50(%)	AP50:95(%)	FLOPs (G)
×	×	×	89.7	83.3	67.4	8.7
✓	×	×	87.2	88.1	70.5	9.5
✓	✓	×	91.4	87.5	73.0	11.1
✓	✓	✓	92.1	87.6	75.0	11.8

89.7% precision. Replacing C2f with our 3CAKC module improves precision to 87.2%, demonstrating the effectiveness of adaptive kernel convolution for feature extraction. The integration of AKSS2D further boosts performance to 91.4% precision and 73% AP50:95, validating the importance of state space modeling for capturing long-range dependencies. Finally, the complete backbone with AKCAttention achieves the best performance of 92.1% precision and 87.6% AP50 with only a modest increase in computational cost.

4.3.2 Neck Component Analysis

Table 5 presents the ablation results for neck components, where we systematically replace C2f modules with 4CAKCMamba and its variants.

The progressive integration of neck components follows a similar trend to the backbone analysis. The 4CAKC module alone provides substantial improvements, while the combination with AKSS2D enables better multi-scale feature fusion. The complete 4CAKCMamba module achieves 92% precision and 88% AP50, demonstrating the effectiveness of our neck design for feature integration and reorganization.

4.3.3 Attention Mechanism Comparison

We conduct a detailed comparison of different attention mechanisms integrated into the YOLOv8 framework on the Railway Pedestrian Detection Dataset, with results presented in Table 6.

Our proposed AKCAttention mechanism achieves the best performance across all evaluation metrics while maintaining competitive inference speed. Compared to the baseline, AKCAttention improves mAP by 1.1% while achieving higher FPS than multi-head attention (MHA) [5] and squeeze-and-excitation (SE) [11] modules. The superior performance of AKCAttention can be attributed to its effective integration of spatial-channel recalibration with adaptive kernel convolution, enabling more precise feature selection without significant computational overhead.

4.3.4 Computational Efficiency Analysis

We analyze the computational characteristics of our proposed modules in Table 7.

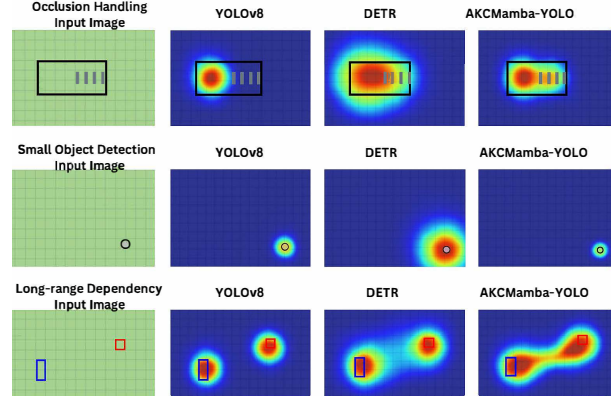


Figure 4. Grad-CAM heatmap comparison of YOLOv8, DETR, and AKCMamba-YOLO under occlusion, small-object, and long-range scenarios.

The analysis reveals that our modules provide excellent performance-efficiency trade-offs. The 3CAKC/4CAKC modules increase parameters by 0.7M but deliver 0.8% mAP improvement. AKSS2D adds another 1.1M parameters but contributes 0.5% additional mAP gain through enhanced long-range modeling. The complete module combination increases total parameters by 2.4M compared to C2f baseline, but achieves 1.6% mAP improvement with only 0.8ms latency increase, demonstrating the effectiveness of our design choices.

4.4. Visualization

4.4.1 Qualitative Visualization and Analysis

To validate AKCMamba-YOLO’s attention distribution, we employ Grad-CAM to generate feature response heatmaps and compare with YOLOv8 and DETR [41] on Railway Pedestrian Detection Dataset, as shown in Figure 4. Visualizations across challenging scenarios including occlusion, small objects, and long-range dependencies demonstrate that our method achieves superior results.

Specifically, under occlusion, AKCMamba-YOLO effectively infers contours of obscured regions to form complete response areas. For small objects, it precisely focuses on information-dense regions within complex backgrounds. In long-range scenarios, the model activates spatially separated objects while implicitly modeling their semantic relationships.

Table 5. Ablation study on neck components using Power Tower Foreign Object Dataset.

4CAKC	AKSS2D	AKCAttention	Precesion(%)	AP50(%)	AP50:95(%)	FLOPs(G)
×	×	×	89.7	83.3	67.4	8.7
✓	×	×	87.4	84.6	70.6	9.3
✓	✓	×	90.8	86.5	71.9	11.6
✓	✓	✓	92.0	88.0	73.3	12.2

Table 6. Comparison of Performance Between Different Attention Mechanisms in YOLOv8.

Model	mAP (%)	AP50 (%)	AP75 (%)	FPS
Baseline	93.2	95.1	73.7	28.6
+ CBAM [33]	93.7	95.6	73.9	27.8
+ SE [24]	93.9	95.8	74.2	27.5
+ MHA [38]	94.1	95.7	74.1	26.9
+AKCAttention	94.3	95.9	74.3	29.2

Table 7. Computational efficiency analysis of proposed modules.

Module	Params (M)	FLOPs (G)	Latency (ms)	mAP (Gain)
C2f (Baseline)	2.1	8.7	2.1	—
3CAKC/4CAKC	2.8	9.3	2.3	+0.8%
AKSS2D	3.9	11.6	2.8	+1.3%
AKCAttention	4.5	12.2	2.9	+1.6%

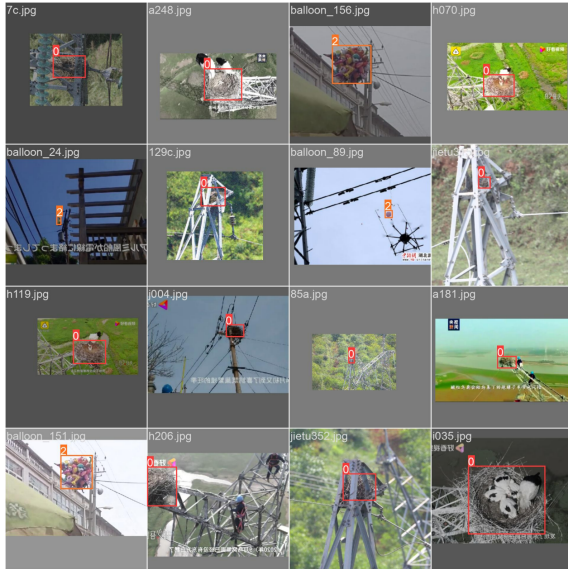


Figure 5. Test on the Power Tower Foreign Object Dataset.

4.4.2 Visualization Results Analysis

As shown in Figure 5, AKCMamba-YOLO demonstrates strong performance on the Power Tower Foreign Object Dataset, effectively detecting subtle anomalies like bird nests and kite strings on power infrastructure. Its precise bounding box regression and accurate classification under

varying conditions highlight the model’s practical value for infrastructure monitoring.



Figure 6. Test on the Railway Pedestrian Detection Dataset.

Figure 6 demonstrates our framework’s performance on the Railway Pedestrian Detection Dataset. The model achieves reliable pedestrian detection under challenging conditions including varying illumination, diverse poses, and significant scale variations. Its consistent accuracy and low false alarm rate confirm suitability for safety-critical applications.

5. Conclusions

We introduced AKCMamba-YOLO, a real-time detector that fuses selective state space modeling with the YOLO architecture through the 3CAKCMamba and 4CAKC-Mamba blocks, which couple AKConv, AKSS2D, and AKCAttention for efficient long-range and multi-scale representation. On COCO2017, the model reaches 46.3% mAP while reducing FLOPs by 47.9% versus YOLOv8-S; it also improves over Mamba YOLO-T. On the Power Tower Foreign Object and Railway Pedestrian Detection datasets, it achieves strong performance in safety-critical scenarios. Future work will explore video detection, edge-device optimization, and multi-modal fusion.

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